

Degradable polylactic acid-reinforced acrylate polymer coating for marine antifouling

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ABSTRACT

Marine biofouling is one of the major challenges facing human exploitation of the oceans. Although marine antifouling coatings are effective in dealing with marine fouling, coatings simultaneously exhibiting low cost, long life, and environmental friendliness are lacking. Herein, a degradable acrylate polymer with long-term effectiveness is prepared; the side chain of this polymer consists of polylactic acid (PLA) and natural rosin. PLA can undergo hydrolysis and degradation on the coating surface, resulting in the antimicrobial and anti-algal properties of the coating while promoting the self-renewal of the polymer-coated surface. The polymer coating thereby exhibits excellent self-polishing properties, with the coating surface maintaining activity even in static environments. The polymer coating exhibits good resistance to protein adsorption as well as excellent bacterial inhibition (>99.9 % inhibition) and algal inhibition (>87.1 % inhibition) because of the synergistic effect between the rapidly degradable PLA and the antimicrobial properties of natural rosin. After 240 days of testing in real marine environments, the polymer coatings still provide good antifouling protection, comparable to that of proven antifouling coatings containing antifouling agents. The proposed coating is an environmentally friendly solution for preventing marine fouling adhesion.

1. Introduction

Marine fouling is one of the major challenges associated with productive activities in the oceans [1,2]. Marine fouling easily occurs on the surfaces of a wide range of underwater structures, increasing the resistance to navigation and fuel consumption of ships [3,4] and accelerating the corrosion of metals [5–7], which lead to considerable economic losses and environmental impacts [8–10]. This has prompted extensive research on marine antifouling materials.

Eco-friendly and efficient marine antifouling coatings have been widely explored and currently include bactericidal [11–13], self-polishing [14–16], biodegradable [14,17,18], micro-structured [19–21], low-surface-energy [21], hydrophilic [22,23], amphiphilic [24], and antifouling release coatings [25]. The generation of marine

fouling is most frequently addressed using self-polishing coatings (SPCs) because of their effectiveness, convenient construction process, and low cost [26,27]. SPCs renew themselves primarily through the hydrolysis of polymer side chains, surface erosion, and shedding [28,29]. However, these processes are highly dependent on the shear forces of seawater flow, and most SPCs exhibit poor polishing performance under static conditions with slow water flow [30,31]. This results in a poor antifouling efficiency of SPCs in static environments.

To address this issue, degradable polymers, such as polycaprolactone (PCL) [2], polylactic acid (PLA) [32], and polyurethane [33], have been widely studied because of their proneness to hydrolysis and environmental friendliness. Among them, PLA has the advantages of degradability, biocompatibility, and renewability of raw materials [34,35]. He et al. reported a thermosetting PLA resin and investigated its

Abbreviations: AA, acrylic acid; AIBN, azodiisobutyronitrile; BA, butyl acrylate; BSA, bovine serum albumin; EA, ethyl acrylate; GPS, gel permeation chromatography; MAA, methacrylate; PBS, phosphate buffer solution; PCL, polycaprolactone; PLA, polylactic acid; SPC, self-polishing coating; TG, thermogravimetric; WCA, water contact angle; XPS, X-ray photoelectron spectroscopy.

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degradability [36]. They reported good biodegradability of the PLA resin and demonstrated the mechanism of PLA degradation in nature, i. e., the final degradation to H_2O and CO_2 through the continuous hydrolysis of ester groups. Darya et al. prepared an environmentally friendly antifouling coating to solve the problem of biofouling in static environments by utilizing the bioactive extract of sea cucumber as an antifouling agent in combination with PCL/PLA [37]. Pang et al. adopted a “from the nature for the nature” strategy to prepare PLA-based polyurethanes with hydrolyzable side groups and produced antifouling coatings with butenolide as the antifouling agent, which is continuously released. However, these coatings quickly degrade, struggling to provide long-term antifouling protection [38]. In addition, antifouling coatings must filled with additional antifouling agents, which is complicated and negatively impacts the environment.

Although natural rosin has traditionally been used as a film former, it exhibits multiple beneficial properties such as environmental friendliness, excellent antimicrobial effect, and antifouling capabilities [39]. We believe it is a favorable alternative to marine antifouling agents. For example, Zhou et al. prepared a natural rosin-based zinc (RZn-x) resin from non-toxic antimicrobial natural rosin through a one-step process, which ensured the green and antimicrobial properties of the material [40]. However, these properties are difficult to balance with the ability to prevent fouling in a static marine environment. Therefore, the simultaneous introduction of rapidly degradable PLA and rosin with natural antimicrobial properties into the acrylic self-polishing polymer is expected to endow the acrylic self-polishing polymer with degradability and excellent antimicrobial properties as well as further improve the polishing performance and antifouling ability of the acrylic self-polishing polymer to realize multifunctional antifouling.

Herein, we present a novel strategy for the preparation of a polymer exhibiting static antifouling properties and long-term effectiveness, suitable for direct application in marine environments without the need for the addition of antifouling agents or fillers. Acrylate polymers with degradable PLA and rosin side chains were synthesized using degradable PLA, rosin acid, ZnO, $Cu(OH)_2$, and acrylate prepolymer as raw materials through a polycondensation reaction. The acrylate backbone structure in this polymer ensures long-term stability during service. In marine environments, the PLA side chains of the polymer are the first to degrade. The resulting lactic acid and fragmented PLA act as antimicrobial agents and effectively inhibit algal growth [41,42]. Subsequently, the O—Zn and O—Cu bonds break, releasing the remaining PLA fragments, natural rosin, and small amounts of Zn and Cu ions (<0.5 %). These components synergistically exhibit excellent antifouling effects. The developed degradable, hydrolyzable environmentally friendly, and long-lasting acrylate polymer lays the groundwork for the development of efficient and clean antifouling coatings.

2. Materials and methods

2.1. Materials

Acrylic monomers (ethyl acrylate: EA, butyl acrylate: BA, acrylic acid: AA, methacrylate: MAA), xylene, n-butanol, and butyl acetate were purchased from Datang Chemical Group Co. Azodiisobutyronitrile (AIBN) was purchased from Tianjin Damao Chemical Reagent Factory. Dodecanethiol was purchased from Sinopharm Chemical Reagent Co. Zinc oxide (ZnO) and copper hydroxide ($Cu(OH)_2$) were purchased from Haisun New Material Co. Rosin resins were purchased from Zhiyuan Chemical Products Co. Lactic acid was purchased from Henan Jindan Lactic Acid Technology Co. Stannous octanoate was purchased from Shanghai Sinopharm Chemical Reagent Co. Agar 2216E and liquid medium 2216E were procured from Haibo Biotechnology Co., Ltd. Seawater was taken from a sea area in Qingdao. All the reagents were used as received without purification.

2.2. Preparation of RPx% polymers and coatings

2.2.1. Preparation of PLA

PLA was synthesized via direct polycondensation reaction, as shown in Scheme 1(a). Under the protection of nitrogen, 50 g of lactic acid and 15 g of xylene were added to a three-necked flask, and azeotropic reflux dehydration was performed at 140 °C for 4 h. Subsequently, 0.25 g of stannous octoate was added to system with temperature controlled at 100 °C, followed by azeotropic reflux dehydration at 180 °C for 20 h to obtain PLA. PLA was characterized using 1H NMR spectroscopy. The degradation experiment was performed by immersion in seawater (Fig. S1).

2.2.2. Preparation of acrylate polymers (RPx%)

RPx% were synthesized following the route shown in Scheme 1(b) and (c). The monomer mixture was composed of 23.9 g EA, 18.7 g BA, 3.4 g AA, 4.0 g MAA, and 0.32 g AIBN (initiator). First, 40 g of xylene, 10 g of n-butanol, and 0.2 g of dodecanethiol were added to a three-necked flask equipped with a stirrer and a reflux condenser tube. Heated in an oil bath, raised to 85 °C, 10 % of the monomer mixture was added and kept at 85 °C for reaction. The remaining monomer mixture was divided equally into 9 portions, which were added to the main system at 15 min intervals. Subsequently, 0.08 g of AIBN was added, and the mixture was maintained at 85 °C for 4 h. Finally, ZnO, $Cu(OH)_2$, rosin, PLA, and solvent were added. For convenience, the resins containing 0, 5 %, 10 %, 15 %, and 20 % PLA (accounting for the molar ratio of rosin) are named RP0, RP5%, RP10%, RP15% and RP20%, respectively, with specific compositions given in Table 1.

2.2.3. Preparation of polymer coatings

ABS boards (8 × 8 cm/10 × 24 cm) were selected as the coating substrate. The surface of the ABS board was sanded, cleaned with ethanol and water, and dried; the surface of the ABS board was planned, and a fixed area of the coating was set; after that, a quantitative amount of polymer samples were taken and brushed on the pre-treated ABS board 2–3 times; after drying naturally for 3 days, it dried in an oven at 45 °C for subsequent tests.

2.3. Structural characterization

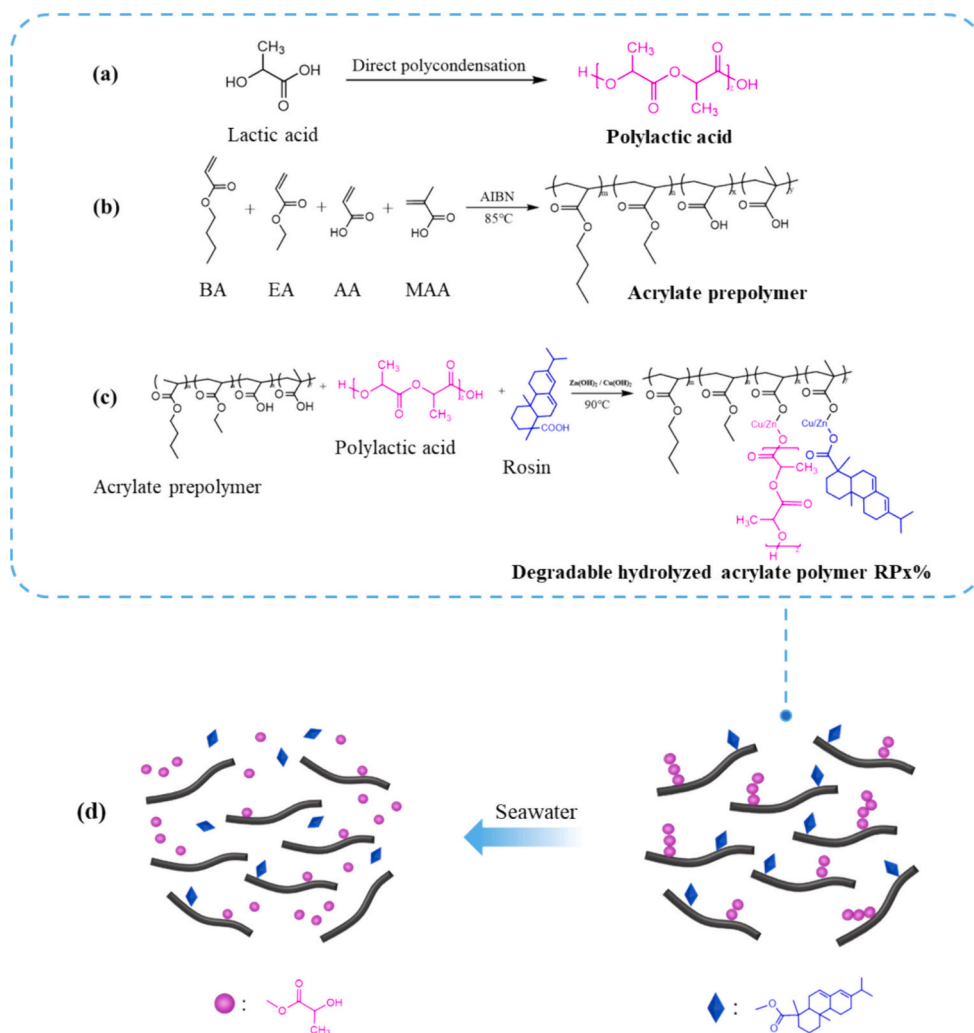
The molecular structures of the prepared PLA molecules and RPx% were determined via 1H NMR spectroscopy (Bruker AVANCE III 600) using dimethyl sulfoxide as a solvent. The molecular structures of the prepolymer and RPx% were analyzed using Fourier infrared spectroscopy (BRUKER, EQUINOX55). The thermogravimetric (TG) properties of RPx% were determined using a TG analyzer (STA449C, NETZSCH, Germany). The elemental distribution of the polymers was analyzed via X-ray photoelectron spectroscopy (XPS; Thermo Scientific K-Alpha). The number-average (M_n) and weight-average molecular weights (M_w) as well as the polydispersity index ($PDI = M_w/M_n$) of the RPx% samples were determined through gel permeation chromatography (GPC; Agilent 1260 Infinity II) using dimethylformamide as a solvent (Table 1).

2.4. Coating surface analysis

2.4.1. Self-polishing performance test

The static and dynamic self-polishing properties of polymer coatings were tested in natural seawater. The prepared polymer-coated samples were subjected to a 7-day cycling experiment in a static or dynamic seawater immersion simulator. At the end of each cycle, polymer-coated ABS panels were removed from the seawater, after which their surfaces were cleaned with deionized water, wiped dry with absorbent paper, and dried in a blast oven at 45 °C for 72 h. Three parallel samples were measured to obtain the average dry weight (W_i). After weighing, the samples were placed in seawater to continue the cycle of experiments.

The rate of mass loss (D) was determined using Eq. (1):



Scheme 1. Synthesis of (a) PLA, (b) acrylate prepolymer, and (c) RPx%; (d) hydrolysis and degradation of RPx%.

Table 1

Composition (molar ratio) and molecular weight of acrylate resins.

Sample	EA/BA/AA/MAA	Rosin	PLA	ZnO/Cu(OH) ₂	M _w	M _w /M _n
RP0	4/4/1/1	1.5	0	1/1	19,357	1.51
RP5%	4/4/1/1	1.5	0.075	1/1	28,698	2.48
RP10%	4/4/1/1	1.5	0.150	1/1	27,577	2.27
RP15%	4/4/1/1	1.5	0.225	1/1	41,505	1.97
RP20%	4/4/1/1	1.5	0.300	1/1	27,262	2.17

M_w and M_w/M_n were obtained through GPC testing.

$$D = (W_0 - W_i)/S \quad (1)$$

where W_0 is the dry weight of the untested coating, and S is the coating area.

2.4.2. Surface observation of the polymer coatings

The changes in the surface morphology of the polymer coatings were observed with a laser confocal microscope (OLYMPUS, OLS 4000). At the end of each cycle of experiments, the sample surfaces were rinsed, carefully wiped, and dried. Afterward, surface observations were performed on samples that underwent static and dynamic marine environment experiments.

2.4.3. Water contact angle (WCA) tests

The surface WCA of the polymer coatings was determined using a contact angle meter (KRÜSS, DSA 100). After the polymer-coated ABS sheet was dried, the initial WCA was determined. The samples were then subjected to cyclic tests on static and dynamic marine environmental test equipment, with 7-day cycles, following a methodology similar to that used by Ni et al. [16]. At the end of a cycle, the samples were removed from the seawater, rinsed, wiped dry, and dried in an oven at 45 °C for 72 h (until constant weight), after which the WCA of the coatings was determined and averaged over three parallel measurements.

2.5. Antifouling experiments

2.5.1. Anti-protein adsorption experiment

First, the polymer coating was pre-soaked in phosphate buffer solution (PBS, pH = 7.2) for 2 h. Afterward, a 0.05 mg/mL bovine serum albumin (BSA) solution was prepared from 20 mg bovine serum albumin and 400 mL PBS. The pre-soaked polymer coating was immersed in 30 mL of BSA solution so that the polymer-coated side of the plate was facing down and floating in the BSA solution for 24 h [43]. After that, the coated surface was rinsed with 10 mL of BSA solution and the washings were collected into the soaking solution. Another 20 mL of PBS was added to the soaking solution. From the above solution, 1 mL was taken and mixed with 10 mL of Kaumas Brilliant Blue working solution; the mixture was shaken well, and its absorbance was measured using UV

spectrophotometry (HITACHI, U-2800, Japan) at 595 nm [44]. The concentration of BSA was calculated from the standard curve (Fig. S2).

The anti-protein adsorption rate (C) was calculated using Eq. (2):

$$C = C_i/C_0 \times 100\% \quad (2)$$

where C_0 is the concentration of unused BSA solution, and C_i is the concentration of protein not adhering to the coating.

2.5.2. Algal adhesion test

Resistance to algal adhesion is a very important capability of antifouling coatings; we tested the resistance of polymer coatings to algal adhesion in natural seawater. *Chlorella vulgaris*, *Skeletonema costatum*, and *Nitzschia closterium*, which are representative algae in marine environments, were selected as experimental algal species. The experiment was conducted as follows. First, the above algae were cultured in filtered natural seawater with added nutrient salts (nutrient salt composition and dosage are provided in Table S1) for 5 days, after which the concentration of the algal solution was adjusted by controlling its 680 nm UV absorbance at 0.1. Subsequently, the polymer-coated samples and the algal solution were placed in the algal Petri dishes and incubated in an incubator at a light intensity of 4000 lx and temperature of 20 °C, with a light-dark cycle of 12:12 h. The algal solution was regularly shaken to ensure the normal growth and adhesion of the algae. After 7 days, the coated surfaces were gently rinsed with 5 mL of seawater to remove unadhered algae, after which the surface of the coated samples was observed for algal adhesion using fluorescence microscopy (Leica, DFC490).

2.5.3. Algal inhibition experiment

In the algal inhibition experiment, the algal species and their culturing were the same as in the adhesion test. After 7 days of incubation, the algal inhibition rate of the samples was calculated as the absorbance of the algae corresponding to the algal concentration. Pure ABS plates were used as control, and three parallel samples were measured to obtain the average value (A_i).

The algal inhibition rate (A) was determined using Eq. (3):

$$A = (1 - A_i/A_c) \times 100\% \quad (3)$$

where A_i is the algal concentration of the coated sample and A_c is the algal concentration of the pure ABS plate.

2.5.4. Antibacterial assay

Three bacteria, Gram-negative (*Escherichia coli*), Gram-positive (*Staphylococcus aureus*), and marine bacteria (*Pseudoalteromonas marina*) were selected for the experiment. All supplies used in the experiment were autoclaved, after which the experiment was conducted on an ultra-clean bench. Bacterial activation was first performed by placing the bacteria in an incubator at 37 °C for 24 h. Afterward, selected strains were inoculated in liquid medium 2216E for 24 h. The solution of cultured bacteria was diluted, and the absorbance of the bacterial solution at 600 nm was adjusted to 0.20. One milliliter of the bacterial solution was taken and dropped onto the resin-coated surface, which was then covered with PE film and sealed in a surface dish. The sample was incubated in an incubator at 37 °C for 24 h. After that, the experimental bacterial solution was diluted with saline to obtain a gradient bacterial solution. The diluted bacterial solution (100 µL) was evenly applied on the dry agar solid medium and incubated at 37 °C for 16–24 h. Pure ABS plates were used as a control, and three parallel samples were measured to obtain an average value.

The bacteriostatic rate (M) was determined using Eq. (4):

$$M = (1 - M_i/M_c) \times 100\% \quad (4)$$

where M is the inhibition rate, M_i is the average number of colonies in the experimental sample, and M_c is the average number of colonies in

the control group.

2.5.5. Antifouling experiments in actual marine environments

The performance of antifouling coatings was tested through real marine environment experiments. Specifically, we conducted marine antifouling experiments on the developed polymer coatings in Qingdao harbor according to the “Method for testing antifouling panels in shallow submergence” (GB/T 5370-2007). We selected PVC board as the experimental substrate material and treated it as follows. The PVC board was sanded with sandpaper, cleaned with ethanol and water, and dried in a natural environment, after which the tested polymer coating was applied with a brush. The samples were brushed twice with an interval of 12 h. The actual marine environment experiments were conducted in an offshore platform where the experimental samples were immersed 1–1.5 m below the sea surface. For observation, the experimental samples were removed, gently cleaned with seawater, photographed and observed, and then quickly returned to their original positions to minimize human influence. The antifouling performance of the coatings in the actual marine environment was evaluated by observing the fouling adhesion on the sample surface.

3. Results and discussion

3.1. Structural analysis

PLA was synthesized via the esterification reaction and tested using ^1H NMR. Fig. 1a shows the ^1H NMR spectrum of the synthesized PLA with the peak at 4.21 ppm attributed to the terminal hydroxyl group of the PLA molecule (a), the peaks at 5.20 and 1.45 ppm were ascribed to the methylene (b) and methyl groups (c) in the molecular chain, respectively, and the intensity ratio of the two typical peaks is 1:3.17. The intensity ratio of the a and b peaks is 1:11.11, indicating that degradable PLA molecules consisting of 11 lactic acid molecules on average were successfully prepared. In the ^1H NMR spectrum of RP10% (Fig. 1d), b and c peaks appear in the same position as the b and c peaks in Fig. 1a. Additionally, in the ^1H NMR spectra of the other RPx% polymers (Fig. S3), protons are observed on the unsaturated carbon of rosin at 5.71 and 5.33 ppm [45]. This indicates that PLA and rosin are present in the synthesized acrylate resin system.

The IR spectra (Fig. 1e, f) show the differences between the acrylate polymers and their prepolymers. As shown in Fig. 1e, a peak at 1603 cm^{-1} is observed in the IR spectra of all the RPx% samples but not in the spectra of the prepolymers. The IR absorption peaks of both COO–Zn and COO–Cu in the polymer system are located near 1603 cm^{-1} [15,46], and the two peaks produce a superimposed peak. The results indicate the entry of PLA and rosin into the resin side chain and the formation of RP10%.

In the TG curves, only a small loss of mass in the samples at the onset (below T1) is observed, which was attributed to the loss of moisture and solvent inside the coating [47]. Above temperature T1, a drastic loss of mass occurs, corresponding to the pyrolysis of the polymer. As shown in Fig. 1g–i, the pyrolysis temperature (T1) of the polymer system decreases with increasing PLA content, with T1 for the RP0, RP10%, RP20% systems being 274.9 °C, 246.1 °C, and 226.7 °C, respectively. At the temperature at which the pyrolysis ends (T2), the mass loss of the polymer also increases with the amount of PLA. This indicates that the introduction of PLA decreases the thermal stability of the polymer. However, the pyrolysis temperatures are all above 225 °C, indicating that the polymers all have good thermal stability (All TG curves are provided in Fig. S4).

The elemental composition of the polymers was analyzed using XPS. Taking RP10% as an example, Fig. 2a indicates the presence of C, O, Cu, and Zn in the polymer. The C 1s peak was used as a reference for the other elements. The atomic content of each element is presented in Fig. 2a, showing that the Cu and Zn contents are very low, 0.31 % and 0.41 %, respectively. In the high-resolution C 1s spectrum (Fig. 2b),

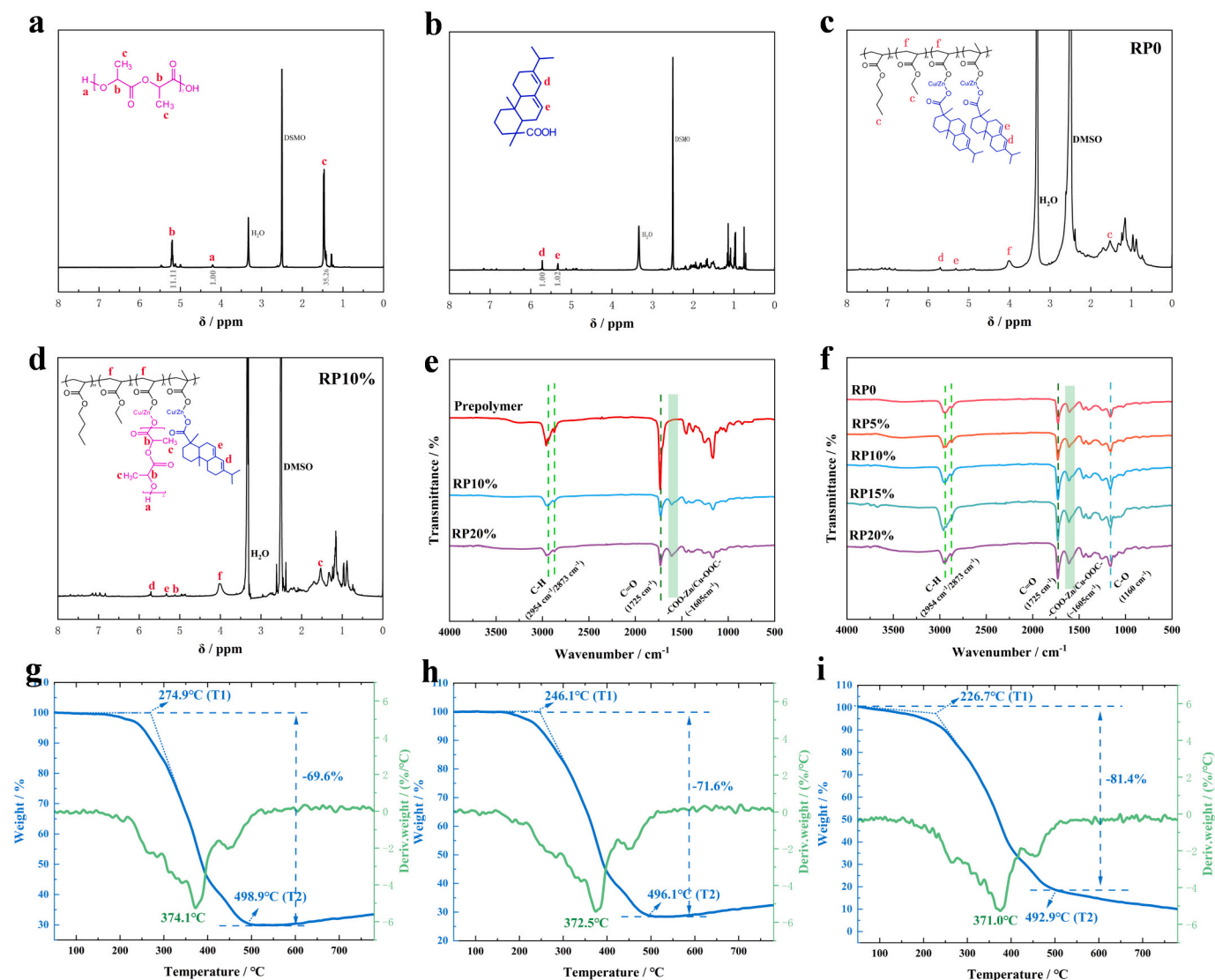


Fig. 1. ^1H NMR spectra of (a) PLA, (b) rosin, (c) RP0, and (d) RP10% (d); IR spectra of (e) prepolymer, RP10%, RP20%, and (f) RPx%; TG curves of (g) RP0, (h) RP10%, and (i) RP20%.

three peaks at 284.8, 286.40, and 288.79 eV correspond to C—C/C=C, C—O, and C=O, respectively [48]. The O 1s spectrum (Fig. 2c) consists of three peaks at 530.35, 532.07, and 533.50 eV, corresponding to O—Me, C—O, and C=O, respectively [49]. The presence of O—Me further confirms the successful grafting of PLA and rosin located in the side chain of the RPx% polymer. In Fig. 2d, two peaks corresponding to Cu $2p_{3/2}$ (933.05 eV) and Cu $2p_{1/2}$ (952.79 eV) are observed [50]. In the Zn 2p spectrum (Fig. 2e), two peaks located at 1022.18 eV (Zn $2p_{3/2}$) and 1045.31 eV (Zn $2p_{1/2}$) were assigned to Zn, confirming the successful synthesis of RPx%, consistent with the results from the ^1H NMR spectra.

3.2. Coating surface analysis

3.2.1. Self-polishing performance test

PLA is highly susceptible to hydrolysis and degradation in the natural environment, forming bioabsorbable lactic acid and its oligomers [51]. To verify the degradation ability of the synthesized PLA in seawater, we performed seawater immersion mass loss experiments in a room temperature environment (Experimental methods in Supplementary Information, Fig. S1). The mass loss after 30 days is 90.6 %, suggesting that the prepared PLA is highly hydrolyzable and degradable. We further

measured the release rates of lactic acid and PLA oligomers from polymer coatings (Fig. S5), and the results demonstrate that PLA in the resin side chain stably released lactic acid and its oligomers, as expected.

We investigated the self-polishing capabilities of the coatings through static immersion in natural seawater and dynamic simulation experiments. As shown in Fig. 3, in both static and dynamic simulation experiments, each coating started to lose weight after immersion in natural seawater, and the weight loss increased with increasing immersion time. Polymer RP0, which does not contain degradable PLA side-chains, shows mass losses of 0.74 and 1.29 $\text{mg}\cdot\text{cm}^{-2}$ after 35 days of the static and dynamic simulation experiments, respectively, which was mainly attributed to the hydrolysis of the ester groups of the side chains in the polymer. In contrast, coatings containing PLA exhibit higher mass loss levels. Particularly, the mass loss levels for the RP20% in the static and dynamic simulation experiments sample are 2.39 and 3.24 $\text{mg}\cdot\text{cm}^{-2}$, respectively.

The results of the degradation experiments show that PLA degrades significantly faster than the O—Zn or O—Cu bonds in the polymer. Consequently, in polymer coatings with PLA side chains, PLA degradation occurs prior to the cleavage of these bonds. The resulting lactic acid or PLA oligomers are released into seawater, accelerating the renewal of the coating surface. At the same time, it is difficult to achieve 100 %

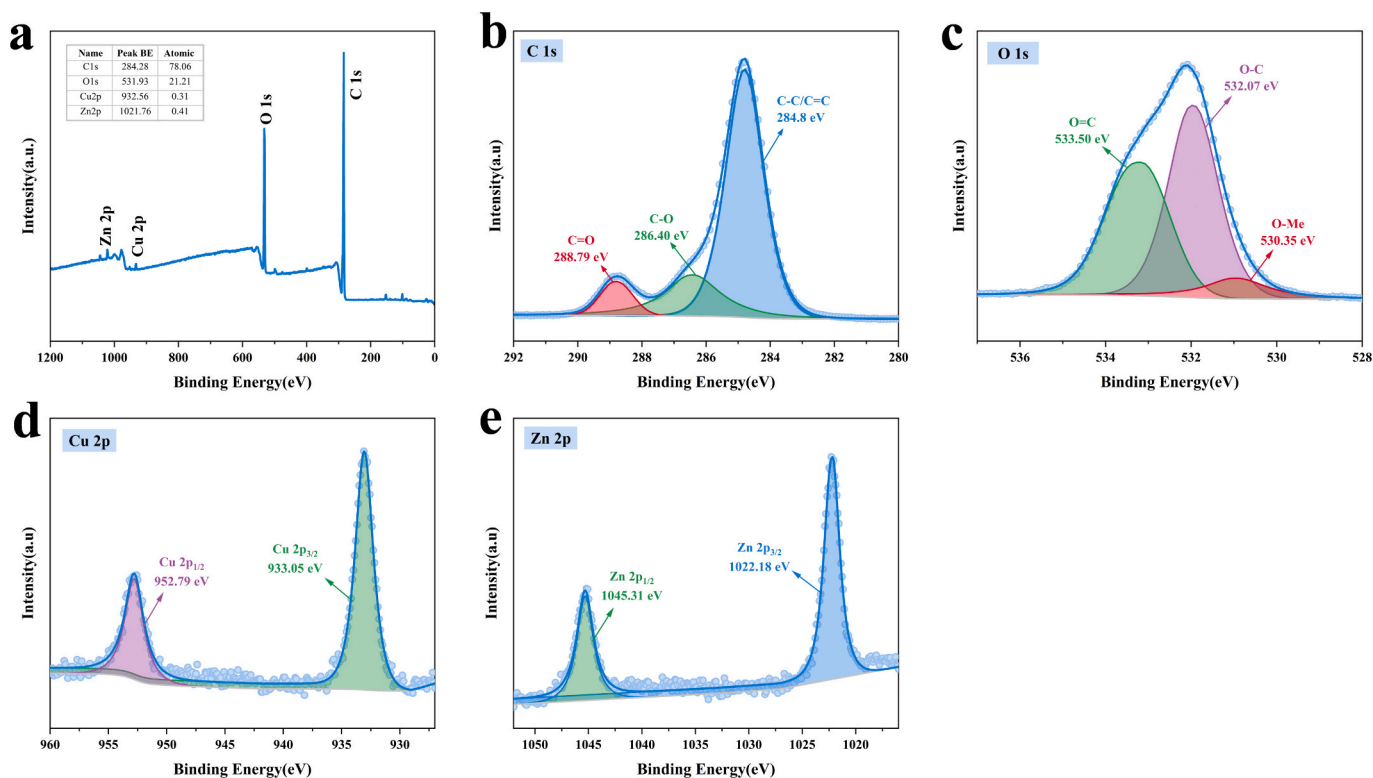


Fig. 2. XPS spectra of RP10% polymers: (a) survey, (b) C 1s, (c) O 1s, (d) Cu 2p, and (e) Zn 2p.

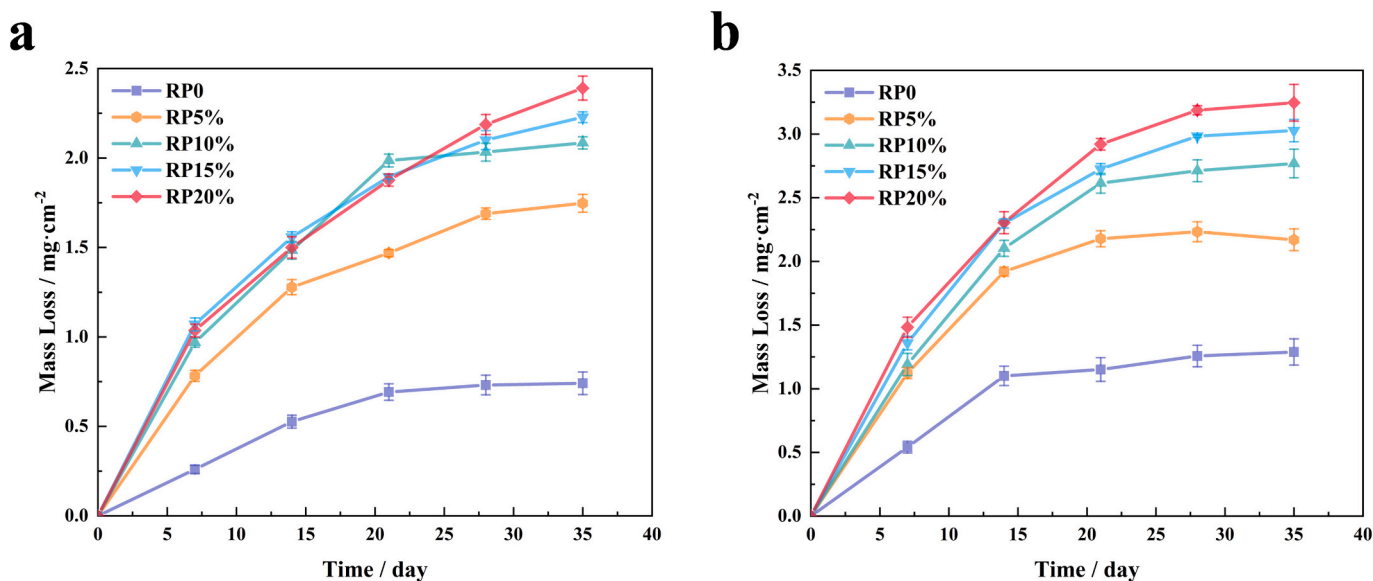


Fig. 3. Time dependence of RPx% polymer mass loss in (a) static simulation experiments and (b) dynamic simulation experiments in room-temperature seawater.

reaction completion with PLA in acrylate resins, so a small amount of PLA is doped into the resin, and the hydrolysis of this small portion of PLA plays a direct role in causing accelerated mass loss.

In addition, the experiment showed the self-polishing rate of the RPx% coatings in static seawater and dynamic seawater is positively correlated with the amount of PLA introduced in the polymer. After 35 days of static and dynamic simulation experiments in seawater, RP0, RP5%, RP10%, RP15%, and RP20% show mass losses of 0.74 and 1.29, 1.74 and 2.17, 2.08 and 2.77, 2.23 and 3.03, and 2.39 and 3.25 $\text{mg}\cdot\text{cm}^{-2}$, respectively. These results indicate that the introduction of PLA accelerates the renewal of the coating surface and that the self-

polishing rate of RPx% can be regulated by the PLA content.

3.2.2. Surface observation of RPx% coatings

As shown in Fig. 4, the surface changes in the RP0 coating are not significant after 35 days of the seawater immersion experiment, indicating the weak renewal ability of the coating. At the same time, significant changes are observed on the surface of the PLA-containing coatings. The degradation rate of PLA itself is significantly higher than the breaking rate of the O–Zn/Cu bonds in the polymer side chains, leading to accelerated renewal of the coated surfaces, consistent with the self-polishing performance.

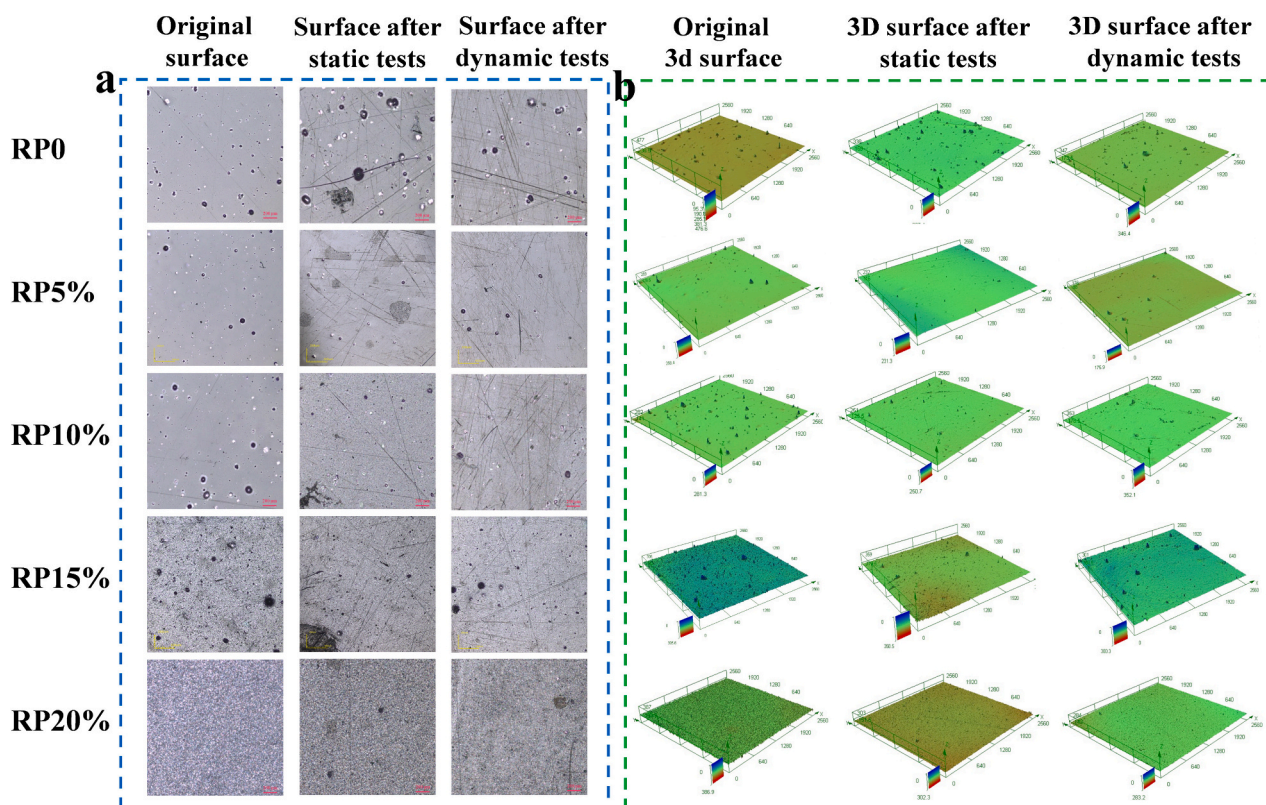


Fig. 4. Changes in the surfaces of RPx% observed using laser confocal microscopy: (a) 2D and (b) 3D images.

3.2.3. WCA test

In Fig. 5, the variation in WCA with time in seawater for the RPx% polymers was investigated during static and dynamic immersion experiments in room-temperature seawater. The coatings all exhibit good hydrophobicity at the initial stage, with WCA around 100° before the simulation experiments. The RP0 coatings show approximately 5° higher WCA, which was mainly attributed to all-rosin side chains. Rosinates contain characteristic bulky hydrophenanthrene ring structures with good hydrophobicity; these structures tend to accumulate in the surface layer, which can significantly change the hydrophobicity of the integrated polymers [52,53].

Overall, in the static and dynamic simulation experiments, the WCA of each coating appears to decrease, increase, and then stabilize. This was attributed to the hydrolysis of the side chains of the acrylate resin, transforming the resin chain segments from hydrophobic to hydrophilic; over time, the hydrophilic coating surface falls off, and a new coating surface is formed [14,26]. The WCA of the RP0 coating changes to a greater extent compared to PLA-containing coatings, especially RP15% and RP20% coatings. This is the case because the surface of RP0 becomes more hydrophilic owing to the hydrolysis of the side chains on the surface, but the hydrophilic coating falls off slower, and the new hydrophobic surface does not appear in time. In the PLA-containing

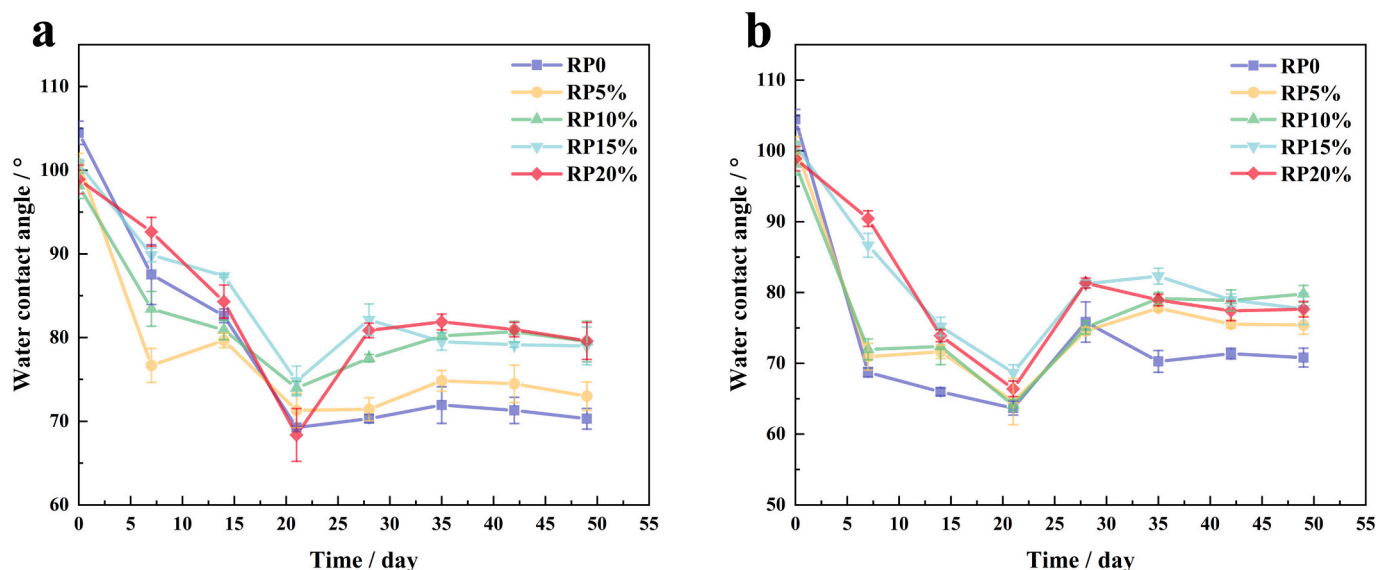


Fig. 5. Variation in WCA with time for the RPx% polymers in seawater: (a) static and (b) dynamic simulation experiments.

coatings, the degradation of PLA accelerates the renewal of the coating surface, so that the hydrolyzed more hydrophilic coating is shed more quickly, and a new hydrophobic surface is formed in time to reach a dynamic equilibrium more quickly, so the hydrophobicity of the coating changes to a lesser extent

3.3. Antifouling performance

3.3.1. Anti-protein adsorption experiment

The attachment of marine fouling organisms begins from the adhesion of proteins to the surface; therefore, the anti-protein adsorption properties of the prepared polymer coatings were determined as shown in Fig. 6. The RP0 coating possesses 76.99 % protein resistance, which was attributed to the protein resistance properties of natural rosin. The introduction of PLA improves the protein resistance of the polymer coatings, with RP15% showing the strongest protein resistance (93.08 %). This was attributed to the rapid renewal capability imparted by PLA to the surface of the coating, making the coating more resistant to protein adhesion. At the same time, the protein resistance of RP20% is lower than that of RP15%. This was explained by the failure of excessive amounts of PLA to react to completion, making the coating surface rough, which is favorable for protein adsorption. The anti-protein adsorption rates of the PLA-containing polymer coatings are all above 80 %, proving that all these resins have certain anti-protein adsorption properties.

3.3.2. Algal resistance test

Algae and bacteria are the main fouling microorganisms. Colonies of these microorganisms produce biofilms that promote the attachment of other marine fouling organisms [54,55]. The excellent algal adhesion resistance of the RPx% coatings is visualized in Fig. 7a. With increasing PLA content, the amount of algae adhering to the surface of the coating significantly decreases, which was attributed to the PLA-enhanced self-renewal of the coatings: the faster the surface renewal, the better the anti-algae performance. Algal inhibition rates of the RPx% coatings were calculated using pure ABS as a control group (Fig. 7b). The inhibition rates of RP0 for *C. vulgaris*, *S. costatum*, and *N. closterium* are 21.8 %, 75.8 %, and 58.7 %, respectively, and the inhibition rates of RP15% are 79.6 %, 84.5 %, and 74.4 %, respectively. With increasing PLA content in the polymer coatings, the algal inhibition of the coatings against the three algal species is enhanced, indicating a positive correlation between the content of PLA in the side chain and algal inhibition. Algal resistance significantly increases with PLA content. RP20% show 87.1 %, 87.2 %, and 89.8 % inhibition of *C. vulgaris*, *S. costatum*, and *N. closterium*, respectively, the best algal inhibition rates among the prepared coatings.

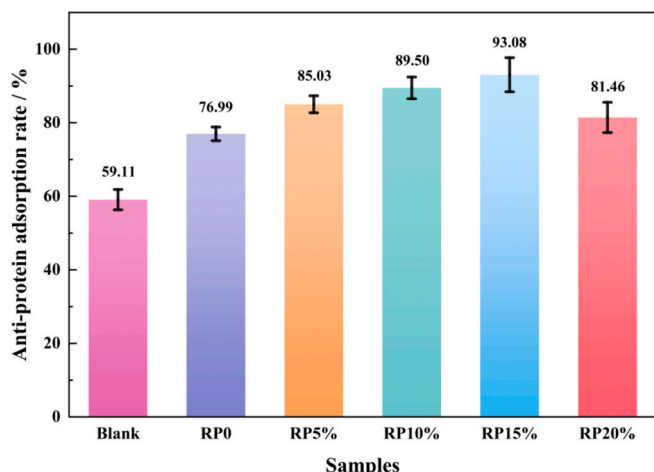


Fig. 6. Protein resistance of the polymer coatings.

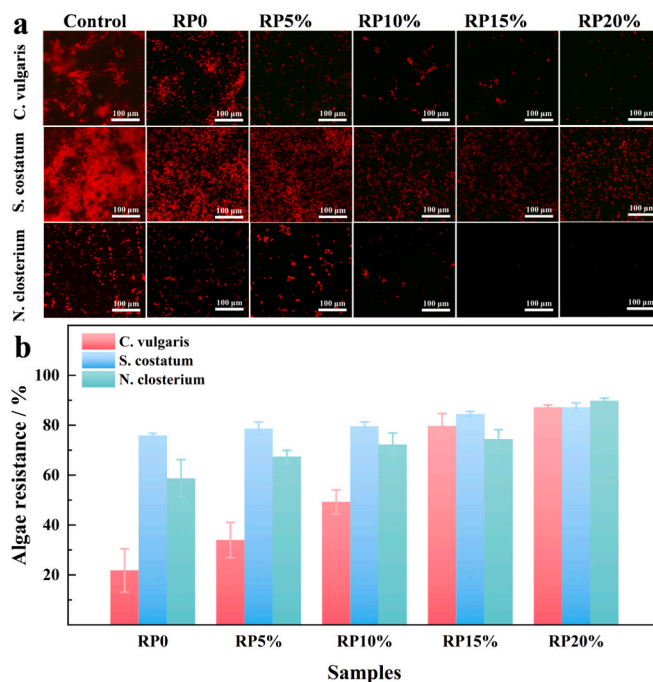


Fig. 7. (a) Microscopic images of the RPx% coatings after the algal adhesion test (scale bar: 100 μ m); (b) algal inhibition rate of the RPx% coatings.

The reason for this result was the inhibitory effect of PLA on the growth of microalgal cells [56]; the sources of PLA include degradation of PLA in the side chains of acrylate resins and the direct release of small amounts of PLA doped into the resin. Algal extracellular polymers (EPS) may increase under the influence of PLA oligomers [57]. EPS can attach to PLA or exfoliated coated particles through the formation of hydrogen bonds or electrostatic interactions and further promote heterogeneous aggregation between algae and particles [58]. Heterogeneous aggregation of algae may lead to direct physical damage, such as altered membrane structure [59] and cell wall damage [60], which may be partly responsible for the inhibition of algal growth by RPx%. At the same time, the combination of heavy metals and microparticles produced by the renewal of the coated surface has been reported to show better inhibition of algal growth than a single substance [28], which is consistent with our experimental results. The inhibition rate of RP20% on all three algae species is >87 %, which is higher than that of the RP0 polymer coating. This suggests that zinc, copper, and lactic acid or PLA oligomers produced by PRx% through hydrolysis and degradation can effectively and synergistically inhibit algal growth.

3.3.3. Antibacterial assay

Bacterial inhibition experiments were conducted using *E. coli*, *S. aureus*, and *P. marina*. As shown in Fig. 8, the RPx% coatings exhibit good inhibition of the three bacteria species, significantly stronger than the RP0 samples without added PLA. The coating show stronger inhibition of marine bacteria, indicating that they are more suitable for application in marine environments. Notably, the antimicrobial effect increases with the PLA content. Notably, RP15% and RP20% show antimicrobial inhibition rates >99.9 % against all three bacteria. This was attributed to the synergistic effect of PLA, zinc and copper metals, and rosin acid groups on the coating surface. Natural rosin exhibits antimicrobial activity [61], providing the main contribution to the antimicrobial effect. Zinc and copper metals affect metabolic processes in bacteria, leading to cellular oxidative stress, which may result in cell damage and sustained apoptosis [62], having a synergistic antimicrobial effect with rosin [40]. At the same time, the degradation of PLA produces lactic acid and PLA fragments, which further enhance the

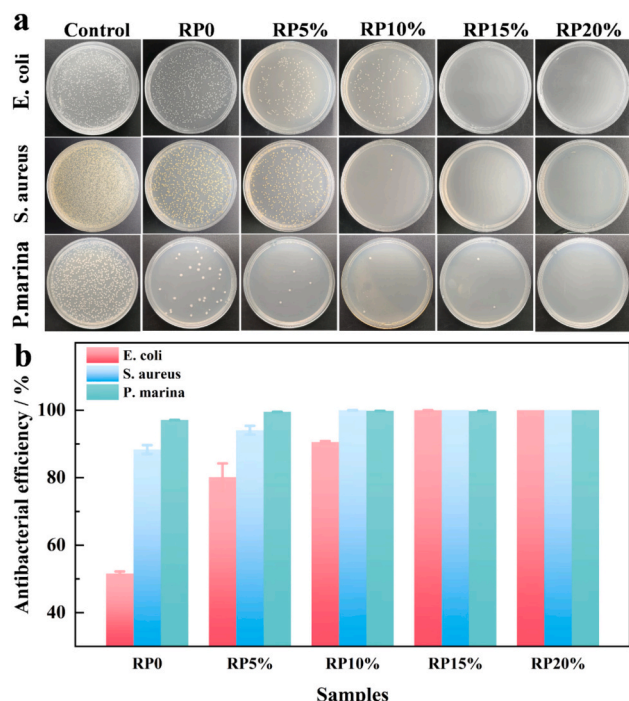


Fig. 8. (a) Inhibitory effects of the RPx% coatings against *E. coli*, *S. aureus*, and *P. marina* (a); (b) bacterial inhibition rates of the RPx% coatings.

antimicrobial effect. Additionally, the rapid degradation of PLA accelerates the dynamic renewal of the coating, so that the antimicrobial groups on the surface of the coating are constantly renewed, contributing to the antimicrobial effect. This makes RPx% effective and long-lasting antimicrobial coatings.

3.3.4. Antifouling experiments in actual marine environments

To assess the practical applicability of the prepared marine antifouling polymer coatings, they were tested in real marine environment. The actual marine environment antifouling test of the RPx% coatings

was conducted in Qingdao harbor waters, with proven antifouling coating containing antifouling agents as a control group (Fig. 9). After 30 days, the blank sample is covered with algae, and bio-adhesion is observed on plates coated with RP0. Silt and algae are observed on blank and RP0 samples after 60 days. In contrast, the polymer coatings containing PLA (RP5%, RP10%, RP15%, and RP20%) all show a very clean surface after 30 days. After 60 days, a minimal number of organisms is observed on the surface of the coatings. After 240 days, polymer coatings RP5%–RP20% still show good antifouling performance. The rougher surface of RP0 was attributed to the lack of PLA in side chains and the slow renewal of the coating surface. The surface of the RP5% sample appears cleaner but still exhibits significant biological attachment. RP10% performs even better, comparable to the control. RP15% and RP20% also show better antifouling than RP0 and RP5%. This suggests that the introduction of PLA does improve the marine fouling resistance of coatings; however, excessive PLA addition may have negative effects. Experiments conducted in real marine environments showed that the addition of moderate amounts of PLA to polymer coatings effectively improves the antifouling performance, even comparable to proven antifouling coatings containing antifouling agents.

4. Conclusion

A degradable and hydrolyzable acrylate resin polymer with PLA and natural rosin side chains was prepared. Possible antifouling mechanisms of coatings were investigated. The acrylate backbone structure of this polymer ensures long-term stability during service. The degradation of PLA proceeds faster than the breaking of O—Zn and O—Cu bonds. Therefore, in marine environments, PLA side chains in the polymer are the first to degrade. This process enhances the surface renewal of the coatings and makes them resistant to the attachment of marine organisms even in static environments. During the initial phase of immersion in seawater, lactic acid and fragmented PLA produced by the coating inhibit microbial activity and algal growth. Subsequently, O—Zn and O—Cu bonds break, releasing the remaining PLA fragments after degradation, natural rosin, and a small amount of Zn and Cu ions (<0.5 %). Rosin plays a major antimicrobial role, while PLA and a small amount of metals (Zn and Cu ions) contribute to the inhibition of algal growth, resulting in an excellent synergistic antifouling effect. The

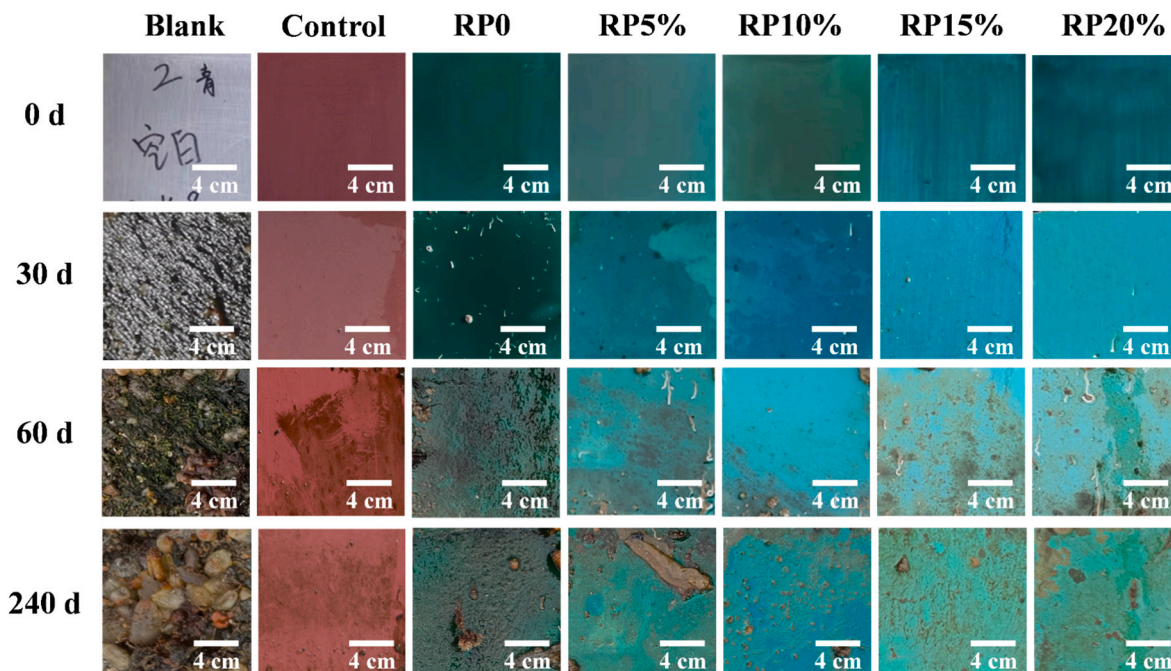


Fig. 9. Antifouling effect of RPx% polymer coatings and control group in actual marine environment.

optimal coating exhibited 87.1 %, 87.2 %, and 89.8 % inhibition rates for *C. vulgaris*, *S. costatum*, and *N. closterium* and >99.9 % inhibition rates for the three considered bacteria. The coatings were tested for 240 days in an actual marine environment and showed good marine antifouling properties, comparable to those of a proven antifouling coating containing antifouling agents. The proposed acrylate polymer is a low-cost, degradable, environmentally friendly, and long-functioning material with promising application potential in marine antifouling and related fields.

CRedit authorship contribution statement

Hao Xu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Yanjuan Wang:** Validation, Supervision, Formal analysis, Conceptualization. **Chunhua Ni:** Writing – review & editing, Formal analysis. **Guangling He:** Writing – review & editing. **Xia Li:** Funding acquisition. **Xiaohui Jiang:** Resources, Project administration. **Liangmin Yu:** Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2024.108824>.

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